1 Two parallel metal plates, each of area A, are separated by a small distance d, as shown in Fig. 1.1.

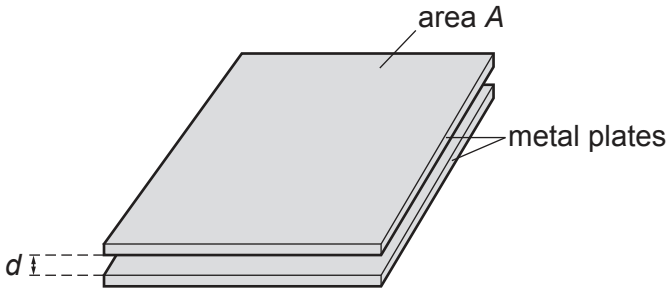

Fig. 1.1 (not to scale)

The plates are initially charged using a power supply.

The plates are then connected to an uncharged capacitor. The potential difference V across the capacitor is measured.

It is suggested that V is related to d by the relationship

$$\frac{W}{V} = 1 + \frac{Cd}{KA}$$

where C is the capacitance of the capacitor, and K and W are constants.

Plan a laboratory experiment to test the relationship between V and d.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for K and W.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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